

Aayush Jagtap

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EDUCATION

Michigan State University

East Lansing, MI

Bachelor of Engineering in Computer Science

Aug. 2023 – Dec. 2026

Courses: Operating Systems, Algorithm Engineering, Data Structures and Algorithms, Software Engineering, Object-Oriented Software Design, Computer Organization and Architecture, Artificial Intelligence

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Rocket Mortgage | Angular, TypeScript, C#, .NET, Jasmine

Detroit, MI

- Developed core UX features for an Angular 17 visual workflow editor in Rocket Logic's loan origination platform, including fullscreen controls, a 7-shortcut keybind system, and a redesigned toolbar
- Resolved a data-corruption bug causing silent data loss across six workflow step types by tracing a nested JSON double-serialization failure, catching a second regression missed by 648 CI checks
- Engineered schema-sourced form defaults via a custom [SchemaDefault] C# reflection attribute consumed by Angular dynamic forms, helping enable 100% of task workflows to be built without raw-JSON expert mode
- Implemented a schema-driven form control override rendering an embedded JSON editor for discriminated-union bypass rules; 50%+ of surveyed users reported standard mode could fully replace expert mode

A.I. Fellow

November 2025 – May 2026

Handshake | Python, AI/ML Evaluation, Data Quality, Multimodal Systems

Remote

- Improved large-scale AI model performance through high-accuracy evaluations of text, image, and audio outputs, maintaining a 96% quality rate across evaluation tasks
- Identified model failure modes and edge cases, delivering structured feedback that informed processing pipelines, alignment logic, and evaluation tooling

Sound Quality Benchmarking Intern

May 2025 – Aug. 2025

Harman International - A Samsung Company | C++, Python, SQL

Novi, MI

- Reduced acoustic testing time by 50% by leading development of a Python GUI automating SoundCheck sequence control via TCP/IP
- Created C++ SQC files with embedded SQL to generate pink noise across targeted car audio channels and capture Frequency Response Function data

Information Technology Intern

May 2024 – Aug. 2024

Camaco-Amvian | Active Directory, Hyper-V, Linux, Microsoft Office

Farmington Hills, MI

- Led virtualization environment setup and VM management with Hyper-V, authoring a report comparing virtualization vs. traditional computing
- Migrated data to the Plex Fixed Asset module in a test environment and demonstrated the process to stakeholders for future rollout

PROJECTS

Branches Project Management System | React, TypeScript, FastAPI, PostgreSQL, Docker, AWS August 2026

- Built a full-stack Trello-style Kanban app with a React + TypeScript frontend, FastAPI backend, JWT authentication, and drag-and-drop ordering persisted to the backend
- Deployed on AWS with the React build served via CloudFront/S3 and a containerized FastAPI backend on EC2 behind Caddy for automatic TLS
- Migrated persistence from SQLite to RDS Postgres with Alembic schema migrations and environment-based configuration, removing all hardcoded credentials

DriveAware Detection System | Python, C, OpenCV, MediaPipe, DeepFace

October 2025

- Developed a real-time driver monitoring system detecting drowsiness and emotional instability via MediaPipe eye-tracking and DeepFace emotion analysis
- Optimized multithreaded processing for low-latency detection and hardware response

SpartanMind Game | C++, GitLab, wxWidgets, XML

April 2025

- Developed XML-based level loading supporting reusable assets, dynamic item placement, and scalable game logic
- Implemented event-driven tile mechanics and authored 20+ unit tests validating gameplay across levels

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, JavaScript/TypeScript, SQL, ARM Assembly, MATLAB, HTML/CSS

Technologies/Frameworks: .NET Core, Node.js, AWS, Docker, PostgreSQL, Linux, Git, FastAPI, React, Angular